

AD 502 – Machine Learning

Pre-Reading Material – Lecture 19

Classification Report

Why This Topic Matters

Accuracy alone does not always tell us how well a classification model is performing. A **classification report** provides a more detailed view of model performance for each class.

The main measures are **Precision, Recall, F1 Score, and Support**.

Learning Outcomes

After completing this pre-reading, you should be able to:

- Explain **Precision, Recall, and F1 Score**.
- Understand the difference between Precision and Recall.
- Explain the meaning of **Support**.
- Interpret a classification report.
- Identify which performance measure is important for a given classification problem.

1. Precision

Precision tells us:

Of all the instances predicted as Positive, how many were actually Positive?

A high precision means the model makes **fewer False Positive errors**.

Think: When the model says "*Positive*," how often is it correct?

2. Recall

Recall tells us:



Of all the actual Positive instances, how many did the model correctly identify?

A high recall means the model makes **fewer False Negative errors**.

Think: Of all the actual Positive cases, how many did the model find?

3. F1 Score

F1 Score provides a combined measure of **Precision and Recall**.

It is particularly useful when we want a balance between:

- Avoiding False Positives, and
- Avoiding False Negatives.

A high F1 Score indicates that the model has a good balance between Precision and Recall.

4. Support

Support represents the **number of actual instances belonging to each class** in the dataset.

For example, if a dataset contains:

- Class A → 60 actual instances
- Class B → 40 actual instances

Then the support for Class A is **60**, and for Class B is **40**.

Support does **not** measure how accurately the model predicts the class; it tells us how many actual samples belong to that class.

5. Classification Report

A classification report typically presents:

Class	Precision	Recall	F1 Score	Support
Class A	Performance of positive predictions	Ability to identify actual class	Balance of Precision & Recall	Number of actual samples
Class B	Performance of positive predictions	Ability to identify actual class	Balance of Precision & Recall	Number of actual samples

It may also provide overall or averaged performance across classes.

Think Before the Lecture

1. What does **Precision** tell us?
2. How is **Recall** different from Precision?
3. Why might F1 Score be preferred over accuracy in some situations?
4. What does **Support** represent?
5. If a model has high Precision but low Recall, what does that tell us about its predictions?

